

Other factors: lead time and demand pattern over the life cycle

Part 7 · Other factors — solution report with data, method, results and charts

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Replenishment lead time

8 days Quoted lead time 65% On-time ($\leq$ quoted)	7.75 days Mean actual 1,550 units ($d = 200/\text{day}$) Pipeline stock at mean LT	1.77 days Std. deviation 10 days 95th-percentile lead time
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Replenishment lead time drives pipeline and safety stock

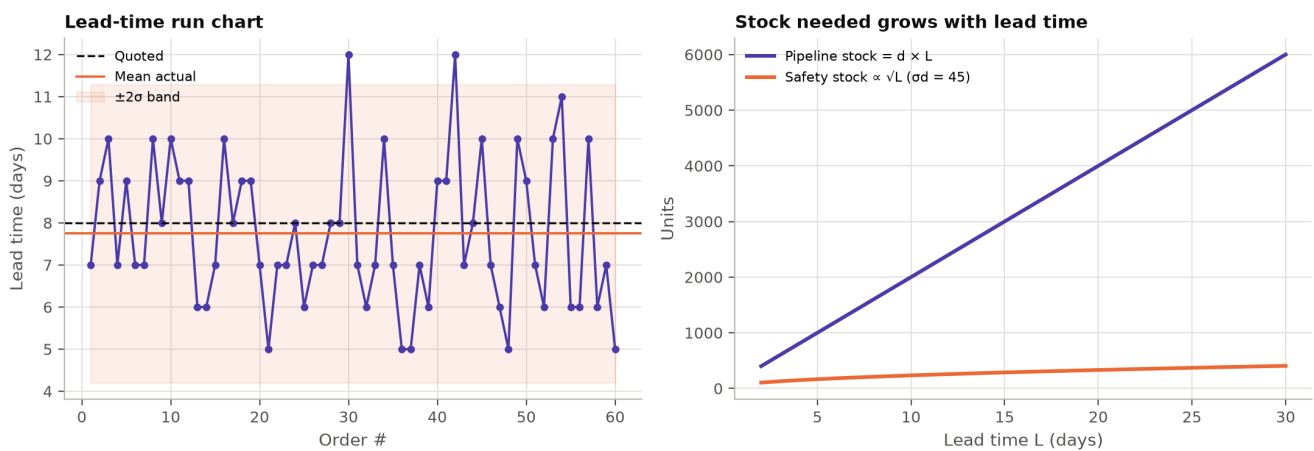


Figure 1.1 — Observed lead times and how pipeline/safety stock scale with the lead time.

Demand pattern and the product life cycle

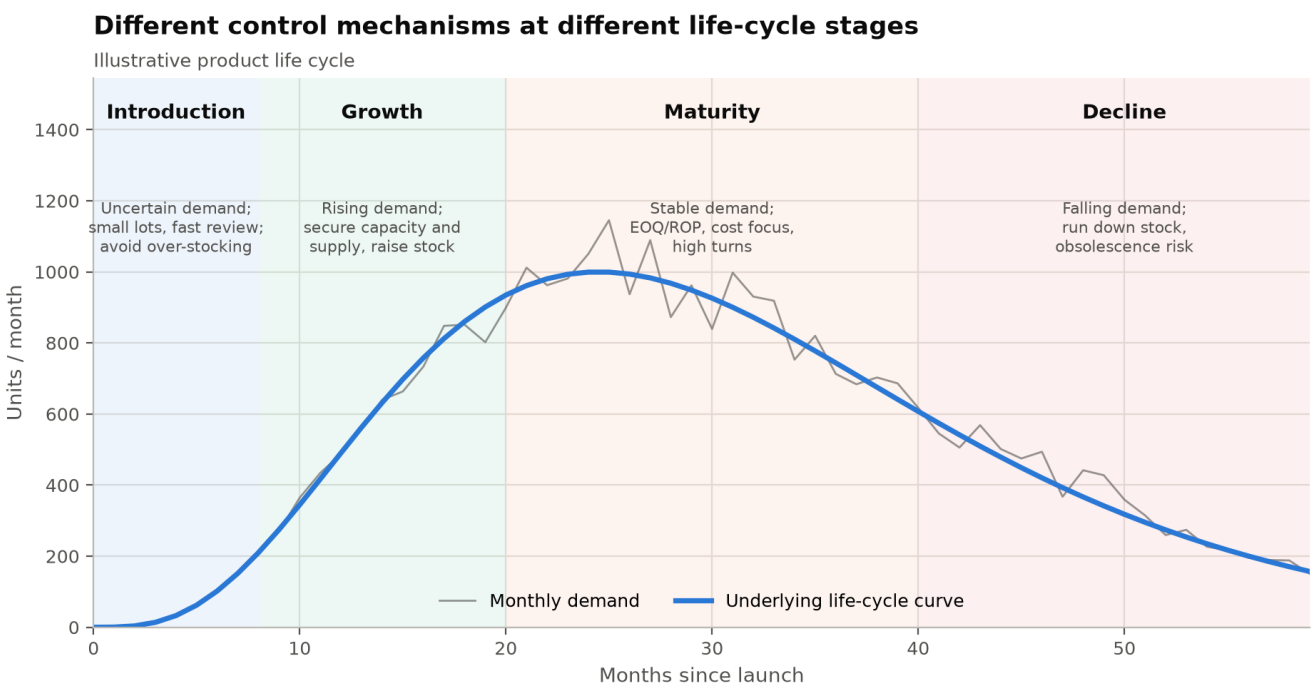


Figure 1.2 — Product life-cycle demand with recommended inventory control emphasis per stage.

Demand statistics by life-cycle stage.

stage	months	avg demand	demand cv	growth per month pct	control emphasis
Introduction	0–8	46	1.14	+45.8	Uncertain demand; small lots, fast review; avoid over-stocking
Growth	8–20	572	0.37	+10.6	Rising demand; secure capacity and supply, raise stock
Maturity	20–40	898	0.15	-2.0	Stable demand; EOQ/ROP, cost focus, high turns
Decline	40–60	366	0.39	-6.7	Falling demand; run down stock, obsolescence risk

Independent vs dependent demand items.

demand type	examples	how determined	typical models
Independent	Finished goods, spare parts, service kits	Forecast from market	EOQ / ROP, periodic review, newsvendor
Dependent	Components, sub-assemblies, raw materials	Calculated from end-item plan × BOM (MRP)	MRP, lot-for-lot, JIT/kanban